

PROFESSIONAL SUMMARY

Computer Engineering student skilled in AI/ML, LLM integration, and backend development. Experienced in building deployed machine learning systems, RAG-based AI assistants, and computer vision applications using Python, Flask, TensorFlow, LangChain, and ChromaDB. Interested in generative AI, machine learning systems, and scalable AI application development.

EDUCATION

Thapar Institute of Engineering and Technology, Patiala, India

Bachelor of Engineering in Computer Engineering | CGPA: 8.57/10 | 2024 – 2028

Manav Public School, Amritsar, Punjab

CBSE Class XII – 89% | 2024

TECHNICAL SKILLS

Languages: Python, C++, C, JavaScript, SQL, HTML, CSS

Machine Learning & AI: TensorFlow, PyTorch, Scikit-learn, OpenCV, Pandas, NumPy

Frameworks & Backend: Flask, REST APIs

Databases: MySQL, PostgreSQL

Developer Tools: Git, GitHub, VS Code, Google Colab

PROJECT EXPERIENCE

INGRES AI ChatBot – Groundwater Intelligence Assistant | [GitHub](#)

(Python, Flask, Google Gemini 2.5, LangChain, ChromaDB, RAG, Leaflet.js)

- Built an AI-powered virtual assistant for **India's Ground Water Resource Estimation System (INGRES)**, enabling natural language querying of official CGWB groundwater assessment data across 6,965+ blocks and mandals.
- Implemented a **Retrieval-Augmented Generation (RAG) pipeline** using LangChain and ChromaDB, ingesting CGWB PDF reports into a vector store for semantically grounded answers with source citations.
- Integrated **Google Gemini 2.5 Flash** with AI-based entity extraction to automatically identify district, state, category, and intent from user queries, triggering dynamic Chart.js visualizations and Leaflet.js maps.
- Added multilingual **support for 8 Indian regional languages** (Hindi, Telugu, Tamil, Kannada, Marathi, Gujarati, Bengali, Punjabi) using Deep Translator and LangDetect for automatic language detection and response translation.

MalariaScope – AI-Powered Malaria Detection System | [GitHub](#)

(Python, TensorFlow, CNN, Flask, Hugging Face, OpenCV)

- Developed an end-to-end deep learning web application utilizing a **Convolutional Neural Network (CNN)** to automate malaria parasite detection in blood smear images.
- Achieved a model **accuracy of 96.8%** and an **F1-score of 0.95** on a dataset of 27,000+ images, significantly reducing the risk of false negatives in diagnostics.
- Optimized image pre-processing using **OpenCV** (resizing, RGB normalization, and noise reduction) to ensure consistent input quality for the **TensorFlow** inference engine.
- Engineered a **Flask backend** to handle concurrent image uploads and served real-time predictions via a cloud-hosted **Hugging Face Spaces** environment.

EcoSense – AI-powered Air Quality Prediction & Analysis System | [GitHub](#)

(Python, Scikit-Learn, Ensemble Learning, Pandas, Matplotlib, Seaborn, GitHub)

- Developed an AI-powered prediction system to estimate the Air Quality Index (AQI) by analyzing environmental pollutants (PM2.5, NO2, O3) and meteorological variables.
- Designed an end-to-end Machine Learning pipeline featuring automated data cleaning, feature scaling, and multi-model evaluation, achieving a high predictive accuracy with an **R2 score of 0.99** for the Random Forest model.
- Implemented an ensemble learning architecture using a **Voting Regressor** (Random Forest & SVR) to ensure model stability, resulting in an **Ensemble R2 score of 0.94** and significantly reducing prediction error (**MAE: 0.08**).
- Built custom logic to map numerical predictions into standardized health categories (e.g., Good, Moderate, Hazardous) and generated feature importance visualizations to identify PM2.5 as the primary driver of air quality degradation.

ACHIEVEMENTS & CERTIFICATIONS

- Machine Learning with Python – Coursera
- Data Structures and Algorithms in C++ – Udemy